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More than objective knowledge: Exploring heterogeneity in individuals' response to a financial education initiative across multiple financial literacy domains

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ABSTRACT

Whereas prior financial education research has focused on increasing *objective financial knowledge*, recent research highlights the importance of also accounting for individuals' *subjective financial knowledge, financial attitudes,* and *beneficial psychological traits* related to personal financial management. However, a holistic perspective of how financial education changes these different domains of financial literacy is lacking, and insights into potential heterogeneity in individuals' response to education interventions is missing. We utilize data from a personal financial management unit at a large Australian university to address this literature gap. We employ Latent Transition Analysis to identify six latent financial literacy states and find that financial education, together with confidence in financial information search, can explain transitions to higher objective and subjective financial knowledge states, more positive financial attitudes, and more beneficial psychological traits related to managing personal finances. We highlight variation in unit response across states, providing practitioners and policy makers with actionable insights.

1. Introduction

For young adults, financial decisions are being made in an increasingly complex environment (Amagir et al., 2022), be it in choosing a phone contract, product insurance, consumer credit, or student loans. These decisions can greatly affect young adults' financial well-being (West and Friedline, 2016). They also face new financial responsibilities from policy change such as the shift from defined benefit to defined contribution pension plans (Hoffmann and Plotkina, 2020) while product innovation such as Fintech apps offering instant credit including "buy-now, pay-later" services make budgeting and financial planning even more important (Schomburgk and Hoffmann, 2023). In many cases, individuals are *compelled* to be financial consumers (e.g., a retirement account is automatically opened for a first or new job), but rarely do these decisions require demonstration of ability to make them. Instead, the required *financial literacy* is typically assumed.

Financial literacy is "a measure of the degree to which one understands key financial concepts and possesses the ability and

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confidence to manage personal finances through appropriate, short-term decision-making, and sound, long-range financial planning, while [being] mindful of life events and changing economic conditions" (Remund, 2010, p. 284). Developing financial literacy is important individually and collectively reflected in the development of national financial literacy strategies, and financial education interventions have been embraced as a means of developing individuals' financial literacy (Fernandes et al., 2014), with increased attention since the global financial crisis (Zhang et al., 2023). However, acquiring financial literacy is costly for individuals with heterogenous expected benefits (Lusardi and Mitchell, 2014). Hence, financial education design should consider such heterogeneity and take into account individuals' "different preferences and economic circumstances" (Lusardi and Mitchell, 2014, p. 31). Related, there are calls for programs to consider targeting and evaluating outcomes beyond financial knowledge, including "perceived financial self-efficacy, willingness to take investment risks, and planning for money" (Netemeyer et al., 2018, p. 85). Each are significant predictors of current money management stress and expected future financial security which explain perceived financial well-being (Netemeyer et al., 2018). Considering these psychological traits and financial attitudes in addition to financial knowledge forms the focus of our paper.

We consider a sample of young adults at a large Australian university. Using surveys of students who enroll in a semester-long personal financial management unit, and a control group who do not, we examine pre- and post-assessments of subjective and objective financial knowledge as well as attitudes and psychological traits related to managing personal finances: financial self-efficacy, propensity to plan for money, financial risk tolerance, and personal saving orientation. The choice of variables is based on research identifying them as influential regarding financial literacy and the effects of financial education (Fernandes et al., 2014).

We use Latent Transition Analysis (LTA) to identify the latent heterogeneity in the sample, using scores on the selected variables to identify latent states (i.e., sub-groups of individuals scoring similarly on the variables of interest) and predict state transitions (i.e., movement between sub-groups from the pre- to post-unit survey) allowing examination of how individuals change over time (Collins and Lanza, 2010). An advantage of LTA is that covariates can provide different types of information or serve different purposes given the two time points. For example, unit participation is used as a correlate of starting states, which allows identifying whether unit enrolment is due to differences in aspects of financial literacy. Unit participation also serves as a predictor of moving between financial literacy states.

We contribute to the existing literature in several ways. From a theoretical perspective, we add to the emerging literature assessing the effectiveness of financial education interventions, which argues that researchers have to account for the role of psychological characteristics besides addressing objective financial knowledge (Fernandes et al., 2014; Netemeyer et al., 2018). While research has started to consider such characteristics in a piecemeal fashion focusing on one variable at a time, we provide a more holistic perspective. This is important, given consumer protection agencies stress the role of psychological factors in financial capability (Consumer Financial Protection Bureau, 2013), and calls for research on how psychological traits influence financial decision-making (Fernandes et al., 2014).

From a methodological perspective, we build on recent literature identifying the dynamics and heterogeneity in financial decision-making (e.g., Hoffmann et al., 2021). Emerging literature uses latent class models to better understand aspects of individuals' financial decision-making (Gerhard et al., 2018) and changes in their financial vulnerability (Hoffmann et al., 2021), but these models have not yet been applied to analyze potentially heterogeneous effects of financial education interventions on different groups of individuals. Our findings build on previous research arguing that financial education programs cannot be evaluated using a "one-size-fits-all" approach (Lyons and Neelakantan, 2008, p. 111).

To preview our results, we identify six distinct financial literacy states that individuals move between over time. Crucially, participation in the financial education unit has a significant association with individuals moving into states of higher objective and subjective financial knowledge, more positive financial attitudes, and more beneficial psychological traits associated with higher levels of financial well-being. Transitions between states are economically meaningful, correlated with improvements in financial behaviors such as keeping track of expenditures and budgeting. Importantly, there is heterogeneity in state transition, depending on individuals' level of confidence in financial information search.

2. Theoretical background

For young adults, becoming financially independent is a key marker of adulthood (Arnett, 2015) and important steps towards "autonomy will be laid down during the college years" (Shim et al., 2010). Evidence from college student samples is important given calls to add financial education units/courses to university curricula (Lusardi and Wallace, 2013) and given the suggested "urgent need" to improve the financial literacy of students (OECD, 2017). Young Australians were identified as a priority cohort in the National Financial Capability Strategy (Commonwealth of Australia, 2022). Financial literacy is about more than "just" knowledge (e.g., Remund, 2010). Indeed, the OECD's International Network for Financial Education (OECD, 2018, p. 4) defines financial literacy as "awareness, knowledge, skill, *attitude* and *behavior* to make sound financial decisions" (emphasis added). Fernandes et al. (2014) also emphasize both the application and ability to perform financial tasks. This view is recognized in national financial literacy strategies using the term "capability" rather than "literacy" to underscore that in addition to knowledge and skills, psychological traits such as financial risk tolerance and self-efficacy should be considered.[1] While we retain the term *financial literacy*, we adopt the broader definition of Fernandes et al. (2014).

[1] Including Australia (Commonwealth of Australia, 2022), the U.K. (Money Advice Service, 2015), and New Zealand (The Commission for Financial Capability, 2021).

Recent meta-analyses evaluating the outcomes of financial education interventions in randomized control trials document "sizable effects on both financial knowledge (0.15–0.2 SD units) and financial behaviors (0.06–0.1 SD units)" (Kaiser et al., 2022, p. 270), in contrast to earlier work by Fernandes et al. (2014, p. 1872) who found much smaller effects. A possible explanation for the variation is the considerable heterogeneity in intervention design and samples (e.g., principles vs. rule-of-thumb designs (Drexler et al., 2014; Skimmyhorn et al., 2016)), participants' income and location (Lusardi, 2002), and educational backgrounds (Corsini and Giannelli, 2021). A general weakness of existing evidence on the effects of financial education "is that it generally does not identify nor isolate the specific components that contribute to particular programs' effectiveness" (Carpena et al., 2015, p. 3).

In our study, we address these limitations and propose a conceptual model presented in Panel A of Fig. 1. Our overall model explicitly allows for heterogeneity in the effectiveness of financial education initiatives across *latent states* of financial literacy. The latent states are reflected in a range of different *indicators* of financial literacy: objective and subjective knowledge, financial attitudes, and beneficial psychological traits. The financial attitudes and key psychological traits we include have been suggested by, among others, Netemeyer et al. (2018), and include financial self-efficacy, financial risk tolerance, the propensity to plan for money, and personal saving orientation. Our objective is not to normatively establish proficiency levels in these characteristics, but rather to document latent heterogeneity in levels of proficiency across states in terms of the mean levels of these characteristics using data collected at two points in time (before and after a financial education unit). We further aim to identify *covariates* that predict state membership and transitions between states, with a particular interest in the role of participation in the financial education unit and confidence in financial information search (Fernandes et al., 2014).

3. Method and data

3.1. Latent transition analysis

To estimate our model, we use latent transition analysis (LTA), an extension of latent class/profile analysis to a longitudinal setting where it is "uniquely positioned to describe stability and change in categorical states across time" (Nylund-Gibson et al., 2023, p. 284). More specifically for our paper, LTA allows: (i) identifying sub-groups of individuals that reflect distinct latent states of financial literacy in a broad sense; and (ii) identifying how individuals transition between these distinct latent states and the covariates that can predict state membership and transitions. While (i) and (ii) are novel and of interest in themselves, our particular focus in (ii) is how transitions are associated with participating in a financial education unit. LTA has been applied to a variety of financial contexts including modeling individuals' heterogeneity and change in financial access (Birkenmaiera and Fu, 2022), household portfolios (Paas et al., 2007), credit use (Białowolski, 2019), financial satisfaction (Brown et al., 2014), and financial vulnerability (Hoffmann and McNair, 2019). The LTA model we employ proposes the evolution in financial literacy states (k) over time t, where the k states are reflected in individuals scores on j indicators, as presented in eq. 1. Eq. 1 specifies the probabilities of the observed data pattern $P(\boldsymbol{Y}_i|\boldsymbol{Z}_i)$ for individual i.

$$P(\boldsymbol{Y}_i|\boldsymbol{Z}_i) = \sum_{k_1=1}^{K}\sum_{k_2=1}^{K}\ldots\sum_{k_T=1}^{K} P\left(\boldsymbol{X}_{i,t=1} = s_1|\boldsymbol{Z}_{i,t=1}\right)$$

$$x\prod_{t=2}^{T} P\left(X_{it} = k_t|X_{i,t-1} = s_{t-1}, \boldsymbol{Z}_{it}\right)\prod_{t=1}^{T}\prod_{t=1}^{J} P\left(Y_{ijt}|X_{it} = \boldsymbol{s}_t\right) \quad (1)$$

In our estimation, we specify six indicators of financial literacy states and use two time periods. Participation in the education unit is the key covariate of the eight covariates, discussed in the following section, to predict initial state membership and transition over time in state membership. The probabilistic nature of this model means that it is not known a priori what the profile of the latent states are or how many exist among the population.[2] Empirical model fit of alternative solutions (Masyn, 2013) identify the number of states. We use the Markov module of *Latent Gold (Version 5)* which uses the Expectation-Maximization (EM) and the Newton-Raphson algorithms (Vermunt and Magidson, 2016) to estimate Eq. 1.

3.2. Measurement scales

The survey included a set of validated scales to measure key variables, which prior research found are either: (i) important in influencing individuals' financial decision-making in general; (ii) help explain individuals' response to financial education initiatives; or (iii) are part of the broader conceptualization of financial literacy advocated above. These variables capture financial knowledge, beliefs in financial ability, financial attitudes, and propensities to plan and save. We distinguish between measures used as indicator variables of participants' state membership, and covariates used to predict starting states and state transitions (i.e., variables

[2] Paas et al. (2007) identify four additional assumptions reflected in models such as equation 1. First, financial literacy indicators, Y_{ijt}, are mutually independent, given state membership of individual i at t – 1 (the local independence assumption). Second, the latent transition structure is a first-order Markov chain, meaning that state membership at time t depends on t-1 but not earlier occasions. In our case with two time periods, this has no practical implication. Third, covariates only indirectly affect state membership, rather than directly affecting the indicators of financial literacy states. Fourth, an individual belongs to only one state at any time period.

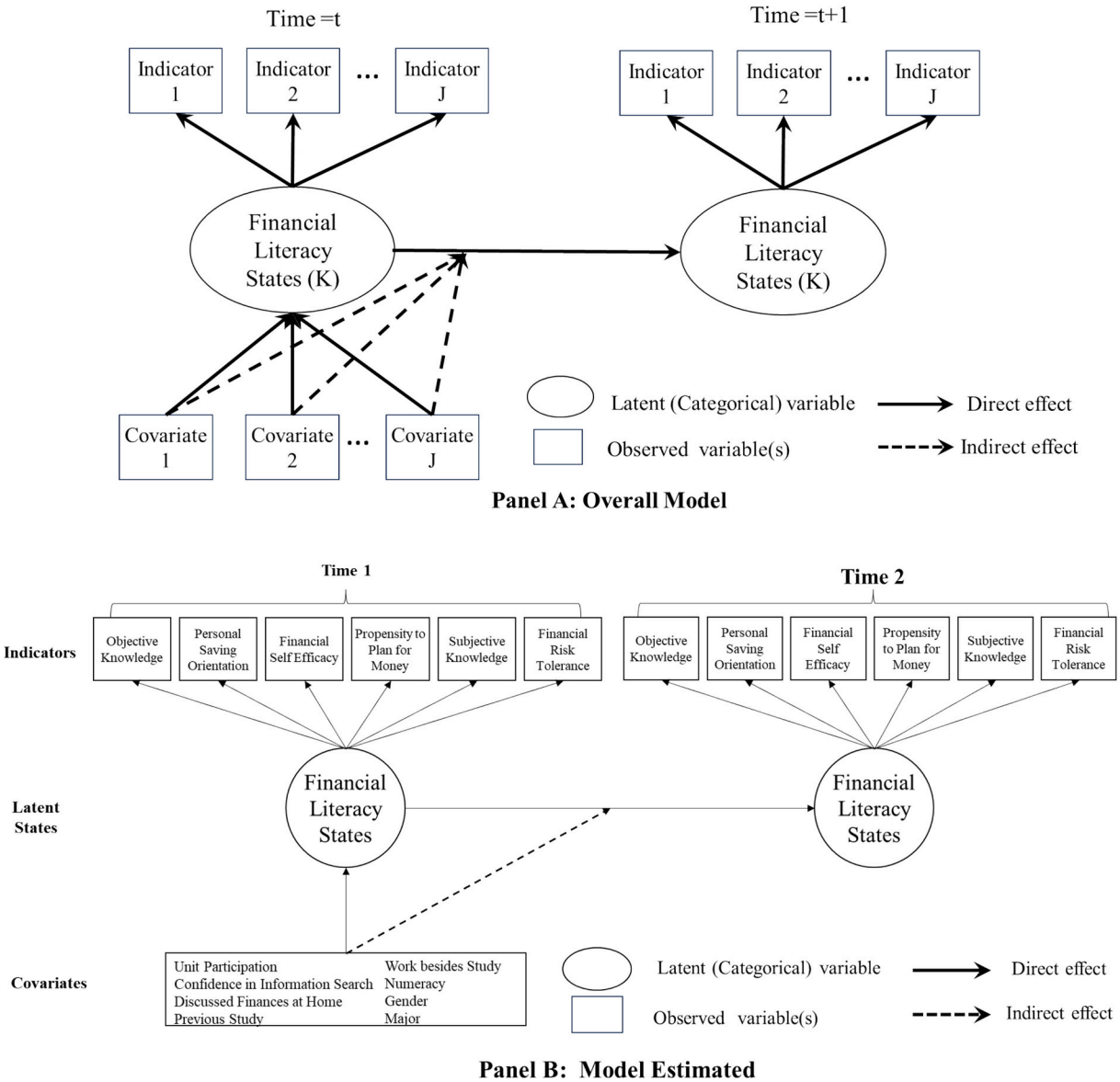

Fig. 1. Conceptual Model.

explaining which sub-group of financial literacy participants belong to and how these change over time).[3] All scales have satisfactory internal reliability, with McDonald's coefficient omega ranging between 0.77 and 0.92 (McDonald, 1999). Further descriptive statistics including scale reliability are presented in Supplementary File Table B.

Indicator Variables of Participants' State Membership.

- *Objective and Subjective Financial Knowledge* using Fernandes et al.'s (2014) 13-item scale for the former, and single item scale of Allgood and Walstad (2016) for the latter.
- *Personal Saving Orientation* assesses the chronic tendency to attach value to saving money in a consistent manner using the 9-item scale of Dholakia et al. (2016).
- *Financial Self-Efficacy* refers to perceived ability of success in managing one's financial affairs, measured by Lown's (2011) 6-item scale.

[3] Please note that the term "indicator" is used broadly here to refer to the group of variables used to infer state membership, in contrast to its narrower use in statistics where it typically refers to a binary or categorical variable.

- *Propensity to Plan for Money* assesses the extent participants proactively manage and plan their financial lives (next 1–2 years), using Lynch et al.'s (2010) 6-item scale.
- *Financial Risk Tolerance* measures willingness to take financial risk and we use the 5-item scale from Jacobs-Lawson and Hershey (2005).

Covariates to Predict Starting States and State Transitions.

- *Unit Participant* indicates participation in the personal finance unit.
- *Discussed Finances at Home* is scored on a single-item scale that gauges amount of discussion of finances at home and acknowledges the financial socialization role of parents (Van Campenhout, 2015).
- *Work Besides Study* is an ordinal scale that measures participants hours of paid work and acknowledges important financial socialization outside of home (Shim et al., 2010).
- *Previous Study* is a categorical variable (high school, previous study since high school, and no previous study) reflecting completion of business, accounting, or economics units, expected to be associated with financial literacy development (Ergün, 2018).
- *Numeracy* is measured using the 3-item formative scale of Schwartz et al. (1997) included as it explains positive financial behaviors (Ghazal et al., 2014) and is positively related to financial literacy (Skagerlund et al., 2018).
- *Gender* is included acknowledging differences in financial literacy between men and women (e.g., Bannier and Schwarz, 2018).
- *Study Major* reflects current study capturing major specific needs (e.g., numeracy).
- *Confidence in Financial Information Search* measures the degree to which participants feel capable and assured with respect to marketplace decisions and behavior, measured with Fernandes et al.'s (2014) 5-item scale. We expect those with more confidence to be more likely to improve their financial literacy over time because of participating in the financial education unit as they will be more open to learning opportunities.
- *Unit Year* controls for potential unit effectiveness differences across time.

3.3. Data collection

Our data comes from an online survey of students enrolled in a first-year elective unit as well as a separate control group of students who were eligible to enroll but did not. The "Managing Your Personal Finances" (MPF) unit is offered as a "broadening" unit at a large Australian university where students are required to complete four "broadening units" from units outside their degree-specific major. The semester-length unit has 36 h of class contact consisting of 12 two-hour lectures and 12 one-hour tutorials. Unit assessment includes examinations, a reflective journal, and an assignment.[4] The unit is delivered within the Finance discipline of the Business School by one of the authors who has been responsible for the unit across the three years we analyze (2017–2019). During these years, the unit's content remained the same. A survey invitation was sent to those enrolled in the week prior to semester. The same invitation was sent to students able to enroll but who did not, to serve as a control group. Survey completion was incentivized by a draw for one of five $50 gift cards. The post-unit survey took place in the week immediately after the end of the unit. The final sample consists of $N = 869$ individuals who completed both the pre- and post-unit surveys, with $n = 535$ enrolled in the unit (treatment group) and $n = 334$ not enrolled (control group).

Comparing Treatment vs. *Control Groups*. As students elect to enroll in the unit, self-selection is a potential source of bias. As described in Section 3.2, LTA estimates both the probability of baseline latent state membership and the role of covariates in explaining state membership at baseline. This allows investigating the association of unit enrolment with baseline state membership. Previewing the results, we find that the Unit Participant covariate is not significant in predicting any of the initial state memberships (refer Table 5). Those who select the unit are not significantly different in terms of latent state membership at baseline.[5]

We also examine differences in covariates by unit enrolment (see Supplementary File Table C, Panel A). There is a smaller proportion of females in the treatment compared to the control group (37.8 % vs. 70.4 %, respectively), consistent with previously-reported gender differences in the take-up of personal finance units (Beierlein and Neverett, 2013). There are no significant differences between the sub-samples by whether finances were discussed at home, confidence in information search, amount of work along with studies, or course major. Numeracy scores are higher (2.4 vs. 2.2) for those enrolled in the unit and the pattern of previous study is different by enrolment. In summary, we find no selection effects in terms of the probability of initial state membership. We do identify differences in some covariates and control for these as all covariates are included in later analyses. However, a limitation remains our inability to control unobservable differences between unit participants and control group.

[4] Students can find information on the hundreds of available broadening units in the annual university handbook. Supplementary File Table A presents the entry for the MPF unit and unit assessment details.

[5] While our focus is on latent states based on the set of indicator variables, Supplementary File Table C- Panel B also summarizes differences in the individual indicators. Unit participants scored lower *Personal Saving Orientation* and *Propensity to Plan for Money* than non-participants, but higher *Financial Risk Tolerance*. Of the latter four traits, only *Propensity to Plan* and *Financial Risk Tolerance* reach a "small" threshold difference with a Cohen's d of 0.21 and 0.30, respectively. All other differences have Cohen's d of less than 0.15 (Cohen, 1988).

3.4. Multiple hypotheses tests

The analysis undertakes multiple comparisons in determining latent states, transition between states, correlates of state membership, and correlates of transition between states. As a result, significant "coefficients may emerge simply by chance, even if there are no treatment effects" (Anderson, 2008), though "the problem is not multiple testing but rather insufficient modeling of the relationship between the corresponding parameters of the model" (Gelman et al., 2012). We control for false discovery rates (FDRs) which seek to control the expected proportion of errors in false rejections (Benjamini and Hochberg, 1995). While FDR has greater power compared to the familywise error rate (FWER), there is a trade-off in accepting a "small number of type I errors in exchange for greater power than FWER control" (Anderson, 2008). All indicators of significance reported in the tables presented in the following section are based on adjusted *p*-values.[6]

4. Results

4.1. Summary of pre- and post-unit change in financial literacy

The full model estimated is presented in panel B of Fig. 1. Table 1 summarizes the differences in participants' scores at the beginning and end of semester across the different indicators of financial literacy. The top panel indicates that there is a significant increase in participants' scores between the pre- and post-surveys within the pooled sample (i.e., both treatment and control group) except for *Financial Risk Tolerance*. The effect size (Cohen's d) is large for *Objective Knowledge* and *Subjective Knowledge* and small for the other measures. For unit participants, there is a significant increase across all domains of financial literacy pre- to post-survey, with a large effect size for *Objective Knowledge* and *Subjective Knowledge* and a small effect size for the other measures. For the control group, the only significant change is for *Financial Risk Tolerance*, which has a decline, though its effect size is small.

4.2. Model fit

We estimate LTA models with varying numbers of states and use the Bayesian Information Criterion (BIC) to identify the model solution with the optimal trade-off between relative model fit and complexity (Collins and Lanza, 2010). Following best practice, we estimate each potential solution using 100 random sets of model parameters to test different potential state profiles and reduce the chance of the model converging on a local instead of a global maximum (Masyn, 2013). Direct effects between indicator variables were included where necessary to satisfy the assumption of local independence (Hagenaars, 1988).[7]

Table 2 summarizes model fit statistics for alternative models with varying numbers of states. Additional latent states were added until the BIC was lowest and additional states resulted in an inferior model fit. The BIC results support a six-state model solution as being optimal, where each of the identified states also have a clearly distinguished profile.

4.3. State profiles

Table 3 summarizes the six-state model solution including state sizes, model parameters, and state profiles. The model parameters represent the relationship between the revealed states and each indicator variable. In practical terms, a significant positive (negative) coefficient for a state-indicator pair indicates that individuals in this state score higher (lower) than the other states on that indicator variable.

State 1 represents about 23 % of the sample. Membership is positively associated with *Objective Financial Knowledge*, *Personal Saving Orientation*, and *Subjective Financial Knowledge*, but negatively associated with *Propensity to Plan for Money*, and *Financial Risk Tolerance*. In this state, individuals have above-average *Objective Financial Knowledge, Personal Saving Orientation, Financial Self-Efficacy, and Subjective Financial Knowledge* but below-average *Propensity to Plan for Money* and *Financial Risk Tolerance*. We label this state '*Knowledgeable*', reflecting its above-average objective and subjective financial knowledge as well as the fact that of its associations with the indicator variables, the positive association with *Objective Financial Knowledge* is most pronounced.

State 2 represents around 22 % of the sample. Membership of this state is positively associated with all indicator variables, showing above-average scores on objective and subjective financial knowledge, positive financial attitudes, and beneficial psychological characteristics. Notably, members of this state score the highest of any state on all indicator variables. Based on these characteristics, we label this state '*On Top of It*'.

State 3 represents about 17 % of the sample and membership is negatively associated with all indicator variables apart from *Personal Saving Orientation,* which is non-significant. The associations with *Objective Financial Knowledge* and *Subjective Financial Knowledge* are most pronounced. This state includes individuals with below-average scores on all indicator variables, including

[6] We use *qqvalue* in Stata (Newsom, 2010) to make the adjustments using option *Simes*. The adjustments do not materially change conclusions based on unadjusted results for the latent state correlates or state transition correlates. Adjusted results retain significance except in a small number of cases where a weakly significant result becomes not significant. For completeness, unadjusted results in Tables D1 to D7 of the Supplementary File which correspond with the same table numbers in the main text.

[7] LTA assumes that indicators within states are locally independent, meaning that the relationships between variables are completely explained by the latent states. We add direct effects between pairs of variables to account for violations of this assumption where required.

Table 1
Pre- vs. Post-Unit Comparisons by Unit Sub-Sample.

		Panel A: Comparison of Indicator Scores Pre vs. Post by State Membership and Unit Enrolment					
		Objective Knowledge	Subjective Knowledge	Personal Saving Orientation	Financial Self-Efficacy	Propensity to Plan Money	Financial Risk Tolerance
Overall Sample							
Pooled							
Sample	Mean	8.10	3.59	5.18	2.62	3.51	3.39
	SD	3.22	1.74	0.89	0.57	1.21	1.16
Pre	Mean	7.20	3.22	5.12	2.58	3.42	3.37
	SD	3.17	1.68	0.91	0.57	1.22	1.11
Post	Mean	9.00	3.97	5.25	2.66	3.61	3.41
	SD	3.01	1.71	0.86	0.57	1.19	1.20
t-test Pre vs. Post		12.13***	9.20***	3.20**	2.88***	3.32**	0.61
Cohen's d Effect Size		0.58	0.44	0.15	0.14	0.16	0.03
Unit Participants							
Pooled							
Sample	Mean	8.71	3.92	5.13	2.64	3.50	3.58
	SD	3.17	1.70	0.87	0.56	1.19	1.14
Pre	Mean	7.33	3.26	5.03	2.56	3.32	3.50
	SD	3.21	1.72	0.89	0.57	1.18	1.12
Post	Mean	10.10	4.60	5.22	2.71	3.68	3.66
	SD	2.44	1.38	0.85	0.54	1.17	1.16
t-test Pre vs. Post		15.89***	14.03***	3.47**	4.30***	5.10***	2.30**
Cohen's d Effect Size		0.97	0.86	0.21	0.26	0.31	0.14
Non- Participants							
Pooled							
Sample	Mean	7.11	3.05	5.28	2.59	3.53	3.08
	SD	3.05	1.66	0.91	0.59	1.24	1.11
Pre	Mean	6.98	3.15	5.25	2.60	3.57	3.16
	SD	3.10	1.62	0.92	0.58	1.25	1.06
Post	Mean	7.23	2.95	5.30	2.58	3.48	3.00
	SD	3.01	1.70	0.89	0.59	1.22	1.15
t-test Pres vs. Post		1.04	1.54	0.83	−0.65	−0.92	1.96
Cohen's d Effect Size		0.08	0.12	0.06	0.05	0.07	0.15
Panel B: 95 % Confidence Intervals for Score Change, Post-Pre by State Membership							
Unit-Participants		2.52–3.02	1.19–1.48	0.12–0.25	0.10–0.19	0.27–0.47	0.07–0.25
Non-Participants		−0.00 - 0.49	−0.35 - -0.04	−0.02 - 0.14	−0.08–0.02	−0.22–0.04	−0.27 - -0.06

Panel A presents a comparison of score means and standard deviation (SD) for key variables at the two time periods for each group. Panel B presents 95 % confidence intervals for score changes by group. Significance is indicated by * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ based on adjusted p-values to account for false discovery rates (Anderson, 2008).

Table 2
Model Fit Statistics.

Model	LL (Log-Likelihood)	BIC (Bayesian Information Criterion)
1 State	−16,873	33,855
2 States	−15,790	32,101
3 States	−15,400	31,937
4 States	−14,821	31,598
5 States	−14,587	32,152
6 States	−13,665	31,532
7 States	−13,685	33,001
8 States	−12,722	32,706

This table presents model fit statistics for models with varying numbers of states which guides the choice of the number of latent states.

objective and subjective financial knowledge, positive financial attitudes, and beneficial psychological characteristics. However, this state does not rate the lowest on any indicator variable of all states. Given these characteristics, we label this state '*Need Help*'.

State 4 represents around 14 % of the sample and is similar to State 3 in that membership of this state is negatively associated with all indicator variables but now the negative association with *Personal Saving Orientation* is also significant. However, the negative associations are even stronger for State 4 than State 3 and most pronounced for *Objective Financial Knowledge* and *Subjective Financial Knowledge*. Individuals in this state score below-average on all indicator variables and the lowest of all states on *Financial Self-Efficacy, Propensity to Plan for Money, Subjective Financial Knowledge,* and *Financial Risk Tolerance*. Given these characteristics, we label this state '*Vulnerable*'.

Table 3
State Profile.

	State 1	State 2	State 3	State 4	State 5	State 6
State Label	"Knowledgeable"	"On Top of It"	"Need Help"	"Vulnerable"	"Need Knowledge"	"Overconfident"
State Size	23.16 %	21.98 %	17.10 %	14.27 %	13.05 %	10.44 %
Model Parameters						
Objective Knowledge	2.06***	3.55***	−1.29***	−1.88***	−2.25***	−0.20
Personal Saving Orientation	0.07***	0.28***	−0.07	−0.23***	0.25***	−0.30***
Financial Self-Efficacy	0.06	0.31***	−0.08**	−0.30***	−0.06***	0.06***
Propensity to Plan for Money	−0.33***	0.48***	−0.37***	−0.53***	0.09	0.65***
Subjective Knowledge	0.53***	2.14***	−1.47***	−2.47***	0.10***	1.18***
Financial Risk Tolerance	−0.28***	0.68***	−0.36***	−0.76***	−0.26***	0.98***
State Profile						
Objective Knowledge	9.70	11.19	6.35	5.76	5.39	7.44
Personal Saving Orientation	5.22	5.43	5.08	4.92	5.40	4.85
Financial Self-Efficacy	2.66	2.91	2.51	2.29	2.54	2.65
Propensity to Plan for Money	3.21	4.02	3.18	3.01	3.63	4.20
Subjective Knowledge	4.00	5.61	2.00	1.00	3.57	4.65
Financial Risk Tolerance	3.13	4.09	3.04	2.64	3.15	4.39

This table provides a profile of the six latent states. The top row indicates the size of each state based on a six-state solution. The next six rows under "Model Parameters" summarize the difference in state mean score from the overall average score for the parameter from a regression of parameter score and state membership. The bottom six rows under "State Profile" summarize the mean scores for each parameter in each state. Significance is indicated by * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ based on adjusted p-values to account for false discovery rates (Anderson, 2008).

State 5 represents about 13 % of the sample and membership is negatively associated with *Objective Financial Knowledge, Financial Self-Efficacy, and Financial Risk Tolerance* but positively associated with *Personal Saving Orientation* and *Subjective Financial Knowledge*. The negative association with *Objective Financial Knowledge* is most pronounced. Individuals in this state score below-average on *Financial Self-Efficacy, Subjective Financial Knowledge,* and *Financial Risk Tolerance* but above-average on *Personal Saving Orientation* and *Propensity to Plan for Money*. However, the most distinguishing feature of this state is that individuals score the lowest of all states on *Objective Financial Knowledge*. Hence, we label this state '*Need Knowledge*'.

Finally, State 6 is the smallest, representing around 10 % of the sample. Membership is positively associated with *Financial Self-Efficacy, Propensity to Plan for Money, Subjective Financial Knowledge,* and *Financial Risk Tolerance* and negatively with *Personal Saving Orientation*. Individuals in this state score lowest on *Personal Saving Orientation and* below-average on *Objective Financial Knowledge*. They score above-average on *Financial Self-Efficacy, Subjective Financial Knowledge,* and *Financial Risk Tolerance*. Given that the above-average subjective financial knowledge of individuals in this state is matched with below-average objective financial knowledge, we label this state '*Overconfident*'.

In Supplementary File Table E, we report post-hoc tests to compare the six states on self-reported past behavior and future

Table 4
Covariates and Starting States.

	Knowledgeable	On Top of It	Need Help	Vulnerable	Need Knowledge	Overconfident	Wald statistic of significant difference
Unit Participant	0.03	−0.45	0.11	0.12	−0.04	0.23	8.70
Confidence in Info Search (t1)	−0.29	1.99***	−1.05***	−1.83***	−0.41**	1.59***	164.67***
Discussed Finances at Home	0.30	0.18	−0.16	−0.10	0.01	−0.23	13.65*
Previous Study							51.87***
No	−0.61**	0.51	0.04	0.50**	0.30	0.28	
High School	0.41*	−0.41	0.29	−0.14	−0.01	−0.15	
After School	0.19	0.92***	−0.34*	−0.36*	−0.28	−0.13	
Work Besides Study							33.28*
No work	0.03	−0.99***	0.40**	0.29	0.75**	−0.48	
<10 h per week	−0.06	−0.17	0.36	0.00	0.44	−0.56	
10–20 h per week	0.20	−0.14	−0.32	−0.22	0.44	0.04	
>20 h per week	−0.17	1.30**	−0.44	−0.07	−1.62*	1.01*	
Numeracy	0.12	0.45*	0.06	−0.26*	−0.17	−0.19	14.25*
Gender (Female)	0.02	−0.88***	0.42***	0.42**	0.24	−0.22	28.18***
Major							32.54**
STEM	−0.20	−0.40	0.17	0.27	0.03	0.13	
Commerce	0.04	1.03***	−0.54*	−0.64*	−0.60*	0.70*	
HASS	0.16	−0.63	0.37	0.37	0.57**	−0.83	

This table summarizes the relationship between covariates and participants predicted starting latent state. The significance of the regression coefficients are based on adjusted p-values to account for false discovery rates (Anderson, 2008). Significance is indicated by: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Unit year (indicators for 2017, 2018, and 2019 are also included but is not significant and not tabulated here.

intentions for four positive financial behaviors from Dew and Xiao's (2011) financial management behavior scale to further validate the meaningfulness of the revealed states, beyond fit statistics. Overall, this analysis confirms the important differences across the latent states as identified by the LTA and demonstrates theoretically plausible associations of these differences with a selection of relevant financial behaviors.

4.4. Explaining starting state membership

Table 4 presents results from the LTA which identifies the role of each covariate on the probability of individuals belonging to each state at baseline. That is, a positive (negative) value for a covariate-state pair represents an increased (decreased) probability of participants belonging to that state based on their score on that covariate. The *Wald* statistic refers to a test of the overall relationship between a covariate and participants' starting state.

Members of the *Knowledgeable* state at baseline are likely to have a lower level of *Confidence in Financial Information Search* and are more likely to have *Discussed Finances at Home* though in both cases the association is relatively weak. Members of this state are less likely to have no previous finance study, and more likely to have studied finance at high school (*Previous Study*). Individuals who were in the *On Top of It* state pre-unit are less likely to participate in the unit (*Unit Participant*), suggesting they recognize their relatively high level of financial literacy, or perceived benefits do not exceed the opportunity cost. Consistent with this observation, they also are likely to have higher levels of *Confidence in Financial Information Search* and *Numeracy* skills. Individuals in this state are also more likely to have continued studying finance *after* high school (*Previous Study*). Individuals in the *On Top of It* state are less likely to have no *Work Besides Study* and are more likely to work more than 20 h per week besides their study. Finally, individuals in this state are less likely to be female and more likely to be enrolled in a Commerce degree (*Study Major*).

Individuals who were in the *Need Help* state pre-unit are more likely to have lower *Confidence in Financial Information Search* and less likely to have studied finance after completing high school (*Previous Study*). They are more likely to not work besides their study. Individuals in the *Need Help* state are significantly less likely to be enrolled in a Commerce degree (*Study Major*) and more likely to be enrolled in a HASS major, though not significantly so. Members of this state are more likely to be female (*Gender*).

Individuals who started as members of the *Vulnerable* state are more likely to have lower *Confidence in Financial Information Search*, more likely to have no previous finance study, and less likely to have studied it after completing high school (*Previous Study*). Members of this state are also likely to have lower *Numeracy* skills, more likely to be female, and less likely to be enrolled in a Commerce degree (*Study Major*).

Individuals who were in the *Need Knowledge* state are more likely to have lower *Confidence in Financial Information Search*, not work besides their study (*Work Besides Study*), less likely to be enrolled in a Commerce degree and more likely to be enrolled in a HASS degree (*Study Major*). Finally, individuals who began in the *Overconfident* state are more likely to have higher *Confidence in Financial Information Search*, work more than 20 h per week (*Work Besides Study*), and be enrolled in a Commerce (*Study Major*).

4.5. Explaining state transitions across time

Overall Transition Probabilities. Table 5 presents overall state transition probabilities, that is the probability of moving between states. Except for the *On Top of It* state, the probability of individuals staying in the same state is less than 50 % for all other states. Transitioning to another state is most common for participants who were originally in the *Need Help* state, as only about 16 % remain in that state at the time of the second survey. Just over 40 % of the members of this state move to the *Knowledgeable* state and about another 13 % transition to the *On Top of It* state. In each of these cases, the transitions represent a shift to states of higher financial literacy. Similar transitions occur for those who were in the *Vulnerable* state, with only a small proportion of individuals remaining in this state over time. Instead, over 30 % move to the *Knowledgeable* state, and about another 12 % move to the *On Top of It* state.

Predictors of State Transitions. We next consider the role of the covariates in transitioning between states, with only the significant covariates reported in Table 6 for clarity. We find that unit participation is associated with state transitions and does so in a beneficial way. Unit participation is associated with a higher likelihood that members of the *Knowledgeable* state transition to the *On Top of It* state. Unit participation is also positively related for transitions from the *Need Help* state to both the *Knowledgeable* and *On Top of It* states. Similarly, being a unit participant is associated with a higher likelihood that members of the *Vulnerable* state transition to either

Table 5
Transition Probabilities.

	State Post					
State (Pre)	Knowledgeable	On Top of It	Need Help	Vulnerable	Need Knowledge	Overconfident
Knowledgeable	35.2 %	44.7 %	7.8 %	4.2 %	1.4 %	6.8 %
On Top of It	6.6 %	82.0 %	1.4 %	0.2 %	2.3 %	7.4 %
Need Help	40.4 %	13.1 %	16.3 %	12.1 %	10.5 %	7.5 %
Vulnerable	30.4 %	12.3 %	12.5 %	31.7 %	4.9 %	8.2 %
Need Knowledge	26.5 %	12.6 %	13.8 %	10.6 %	27.5 %	9.0 %
Overconfident	11.1 %	30.2 %	8.0 %	2.3 %	4.0 %	44.5 %

This table summarizes the overall probability of transitioning from a latent state to another. For example, the top row indicates that 35.2 % of those who begin in the Knowledgeable state remain in the Knowledgeable state at the second period.

the *Knowledgeable* or *On Top of It* states. The likelihood of members of the *Need Knowledge* state transitioning to either the *Knowledgeable* or *On Top of It* states is also positively associated with Unit participation. Unit participation is furthermore positively associated with the transition from the *Overconfident* state to the *On Top of It* state.

Fig. 2 graphically summarizes the size of the differences presented in Table 6. For each starting state, the horizontal bar represents the proportion of individuals who transition to each ending state. The blue-shaded bars are those that are relatively more positive ending states whereas the red shaded bars are those that are relatively more negative. The green bar represents the *Overconfident* state, comprising both positive and negative elements. The top panel indicates that, compared to the control group not taking the unit, unit participants transition to (or remain in) more positive states, as for every starting state the blue shaded bars form the majority. The exception is the *Overconfident* state, where the majority remain in this state, though those that do transition do so to a more positive state. The lower panel for the control group not taking the unit indicates that these individuals predominantly remain in their starting state. Notably, the red-shaded cells remain large at the second period. Each of these changes suggests that participation in the personal finance unit is associated with individuals moving from states of lower to higher objective and subjective financial knowledge, positive financial attitudes, and beneficial psychological traits.

Apart from *Unit Participant*, *Confidence in Financial Information Search* also has a significant association with state transitions. Notably, individuals who score higher on *Confidence in Financial Information Search* are less likely to transition out of the *Knowledgeable* state to the *Overconfident* state. Moreover, higher *Confidence in Financial Information Search* is positively associated with those in the *Need Help* state moving to the *On Top of It* state, while it is negatively associated with these individuals moving to the *Vulnerable* state. Higher *Confidence in Financial Information Search* is also negatively associated with individuals in the *Need Knowledge* state transitioning to the *Vulnerable* state as well as those in the *Overconfident* state transitioning to the *Need Help* state. Hence, our results suggest *Confidence in Financial Information Search* to be an important mechanism facilitating transitions to states with higher objective and subjective knowledge and positive psychological characteristics.

Other covariate associations are significant, although with smaller magnitudes. Among these, females are less likely to transition from the *Knowledgeable* and *Need Help* states to the *On Top of It* state, while females are more likely to transition from the *Overconfident* to the *On Top of It* state and the *Need Help* to the *Vulnerable* state. Finally, individuals with higher *Numeracy* are more likely to transition from the *Vulnerable* to the *On Top of It* state and less likely to transition from the *Knowledgeable* to the *Need Help* state, and the *Need Help* to the *Need Knowledge* and *Overconfident* states, respectively.

Table 6
Covariates and State Transitions.

		State (Post)						
	State (Pre)	Knowledgeable	On Top of It	Need Help	Vulnerable	Need Knowledge	Overconfident	Wald statistic of significant difference
Unit Participant	Knowledgeable		8.77**	−3.00	−2.01	0.12	8.34*	111.59***
	On Top of It	−1.74		1.04	−0.16	−3.14	1.34	
	Need Help	1.36***	4.78*		−2.61	0.68	0.79	
	Vulnerable	3.10***	11.98*	1.57*		0.28	4.60*	
	Need Knowledge	6.94**	5.02*	−0.71	−0.29		6.81*	
	Overconfident	0.29	8.22*	0.61	1.86	2.00		
Confidence in Financial Info. Search	Knowledgeable		−1.80	0.30	−2.20	−1.33	−5.49*	52.79**
	On Top of It	−0.97		−2.14	−1.97	−4.42	1.08	
	Need Help	0.26	1.83		−1.45*	0.77	0.51	
	Vulnerable	0.01	1.22	0.03		2.05	3.33*	
	Need Knowledge	−2.32	−1.75	−0.80	−0.97		−2.77	
	Overconfident	0.14	0.18	−4.69	−2.88	−4.34		
Gender (Female)	Knowledgeable		−4.11**	2.40	−3.75	−1.93	−2.49	58.37**
	On Top of It	0.50		0.78	0.87	0.66	−1.08	
	Need Help	−0.44	−2.05*		1.28*	1.30	0.85	
	Vulnerable	0.37	−0.46	0.09		0.01	−1.13	
	Need Knowledge	0.69	−0.84	0.44	1.93		−0.51	
	Overconfident	3.25	4.28*	−2.00	1.60	1.88		
Numeracy	Knowledgeable		0.35	−3.71	−0.42	−0.49	−1.30	50.23*
	On Top of It	−0.41		−0.65	−0.57	0.43	−0.77	
	Need Help	0.72	−0.44		−0.02	−2.12**	−1.95*	
	Vulnerable	0.20	4.40*	−0.03		0.06	−0.25	
	Need Knowledge	3.29*	2.42	0.40	0.79		7.75*	
	Overconfident	1.10	1.47	1.41	−0.13	1.93		

This table expands on the results in Table 5 by summarizing the relationship between covariates and the probability of transition between states, The significance of the regression coefficients is based on adjusted p-values to account for false discovery rates (Anderson, 2008).A Wald test of the joint significance of a covariate is also presented in the final column. Significance is indicated by: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Additional covariates *Discussed Finances at Home, Previous Study, Work Besides Study, Study Major,* and *Unit Year* dummies were included in the analysis but are omitted from the table for clarity as the Wald test indicates they are insignificant. They are significant.

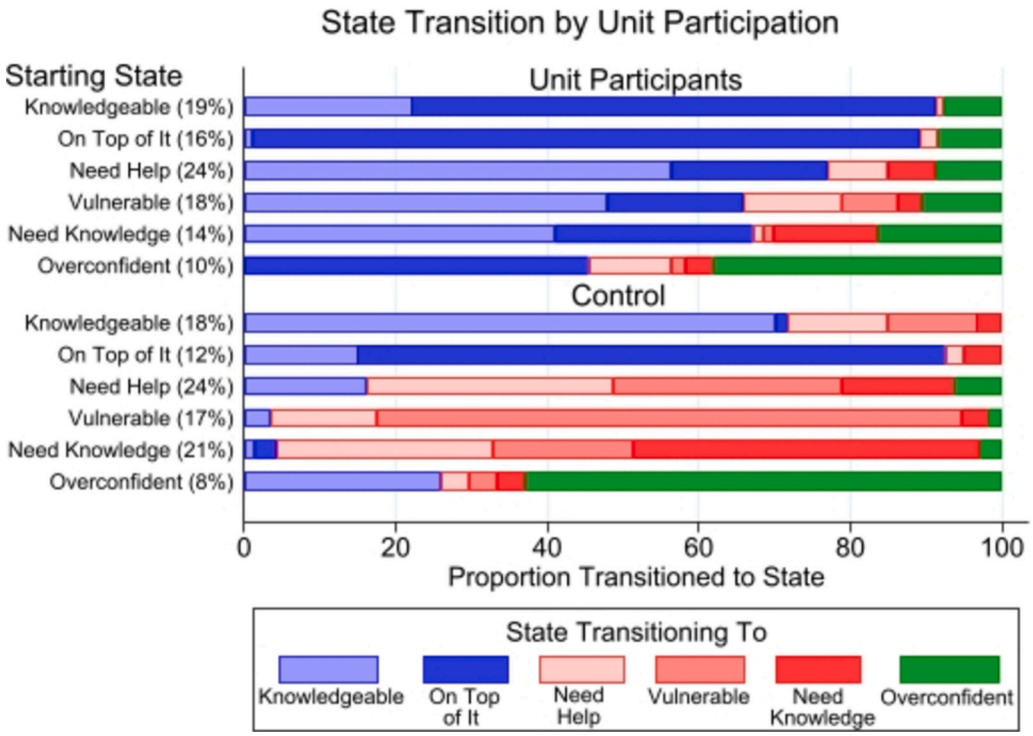

Fig. 2. State Transition by Unit Participation.

4.6. *State transitions and behavior change*

Just as we expect certain positive financial behaviors to be correlated with individuals' starting state membership, we expect state transitions to be correlated with behavior change. As an extension of the post-hoc analysis in Supplementary File Table E, Supplementary File Fig. A summarizes the direction of change, showing that the frequency of participants displaying positive financial behaviors increases as they move out of the problematic states.

5. Discussion of results

5.1. *Six characteristic state profiles*

Our analysis identified that a six-state model best fits the financial literacy data, articulated via objective and subjective financial knowledge, financial attitudes, and beneficial psychological characteristics. Two clearly more positive states emerge, *On Top of It* and *Knowledgeable*, which differ mainly in their *Propensity to Plan for Money* and *Financial Risk Tolerance*. Individuals in the *On Top of It* state also differ from all other states in terms of higher *Financial Self-Efficacy*. Similarly, two clearly more negative states emerge, *Need Help* and *Vulnerable*, which whilst ostensibly similar, vary in the severity of their deficiencies. Whereas individuals in the *Need Help* state have below-average scores on all indicator variables, individuals in the *Vulnerable* state not only score below-average on all indicator variables, but also score lowest of all states on *Financial Self-Efficacy, Propensity to Plan for Money, Subjective Financial Knowledge,* and *Financial Risk Tolerance*.

Regarding the role of the covariates on individuals' starting state membership probabilities, our analysis provides three key insights. First, having *Confidence in Financial Information Search* protects individuals from being in the *Need Help* or *Vulnerable* states; helps them be in the *On Top of It* state; but can also make individuals more likely to be in the *Overconfident* state. This insight builds on prior research finding that *Confidence in Financial Information Search* is positively associated with beneficial financial behaviors (Fernandes et al., 2014) and taking financial responsibility (Ward and Lynch, 2018). Second, *Numeracy* skills facilitate individuals' membership of the *On Top of It* state and help prevent them from being in the *Vulnerable* state, highlighting the value of general numerical ability in supporting individuals' financial literacy. This insight builds on Carpena and Zia (2020) who propose a broader definition of financial knowledge which includes numeracy skills as well as financial awareness and positive attitudes to personal finance. Third, we find an important role for gender: being female is associated with a reduced likelihood of being in the *On Top of It* state, while it increases the likelihood of being in the *Need Help, Vulnerable,* and *Need Knowledge* states. These findings build on early work finding that, compared to men, women score lower on financial literacy (Lusardi and Mitchell, 2011) and self-efficacy (Furrebøe and Nyhus, 2022).

Our analysis also shows behavioral differences across all states regarding ongoing financial planning, including short-term (i.e. budgeting, tracking spending) and longer-term behaviors (i.e. having financial goals). Notably, we find that the most significant changes in financial behaviors over time occur for those in the *Vulnerable* state, and are predictive of transitions of these individuals to more positive financial literacy states. Kaiser and Menkhoff (2017) document minimal effects of financial education for low-income individuals, whereas Kaiser et al. (2022) find no significant role of income in attenuating education outcomes. Our results extend both findings and suggest that income and financial literacy are not substitutes when trying to understand impacts of education interventions. We document that the financial education unit has the strongest association with individuals moving to more positive financial literacy states for those in the *Vulnerable* starting state.

5.2. Changes in state membership following the financial education unit

Key to our Latent Transition Analysis was detecting the probabilities with which members of each financial literacy state moved to alternative states over the time period of the financial education unit. Overall, the likelihood of remaining in a starting state in terms of financial literacy profile was below 50 %, except for those in the *On Top of It* state, who are very likely to remain in that state (82 %), which is of course desirable. It is promising that more than 70 % of those in the *Need Help* state transitioned to more positive states following the education unit while almost 70 % of those in the *Vulnerable* state did so. Beyond expected improvements on *Objective Financial Knowledge* and *Subjective Financial Knowledge*, all unit participants showed significant improvement in terms of their *Personal Saving Orientation*, *Financial Self-Efficacy*, and *Propensity to Plan for Money* (Table 1). In light of recent research indicating that these psychological characteristics mediate the link between financial vulnerability and negative financial outcomes (Hoffmann and McNair, 2019), our results highlight these as worthwhile targets for financial education initiatives.

Notably, those in the *Overconfident* state are the second most likely to remain in their starting state (44.5 %), highlighting that miscalibrations in financial knowledge might be difficult to ameliorate. The *Overconfident* classification reflects a type of overestimation (Inghelbrecht and Tedde, 2024) inferred from subjective and objective knowledge scores. A challenge remains in reinforcing good behaviors of those in this state while guarding against the overconfidence which can lead to poor financial behaviors including excessive trading (Barber and Odean, 2000) and under-diversification (Broekema and Kramer, 2021). Given they overweight their own information and discount conflicting information (Inghelbrecht and Tedde, 2024), a lever for financial education initiatives may be to instill more natural "devil's advocate" tendencies in the overconfident (Arkes, 2001). However, further evidence of the effectiveness of broad financial education interventions on debiasing is needed (cf. Jugnandan and Willows, 2023).

Our findings also show that those with higher *Confidence in Financial Information Search* were more likely to transition to more positive states of financial literacy. These results demonstrate that "soft" skills, such as reassurance in applying financial knowledge, should be as central to financial education initiatives as building factual and practical knowledge. Such skills can be important drivers of financial inclusion (Mindra et al., 2017) and taking financial responsibility (Ward and Lynch, 2018).

Finally, we observe that females are less likely to transition from negative financial literacy states and are indeed more likely to move into an even more negative state. These results build on Hoffmann et al. (2021), who find that compared to men, women are more likely to experience an increase in financial vulnerability over time, which is concerning.

6. Conclusions

To date, objective knowledge measures of financial literacy dominate the literature (Fernandes et al., 2014). While attractive, and efficient, in the context of national surveys (Lusardi and Mitchell, 2011) these measures are reductive (Ranyard et al., 2020). Our findings support the notion that a more holistic approach to financial literacy assessment is useful in evaluating the role of financial education. In particular, our findings highlight that measures of *Subjective Financial Knowledge* and *Confidence in Financial Information Search* are useful means of assessing the role of financial education interventions. Further, our findings support the inclusion of psychological characteristics such as *Personal Saving Orientation* and *Financial Self-Efficacy* into the conceptualization and assessment of financial literacy.

Our contributions are theoretical as well as methodological. Theoretically, we offer a more nuanced understanding of how financial knowledge, financial attitudes, and psychological characteristics interplay with respect to individuals' financial literacy. Our work highlights that financial literacy can, and arguably should, be conceived of as more than mere factual knowledge, with traits such as *Personal Saving Orientation* and *Propensity to Plan for Money* as indicators distinguishing different "states" of financial literacy. Methodologically, we stress the utility of dynamic latent class models to not only measure changes in literacy states, but also pinpoint where heterogeneity underpinning changes in characteristic states exists.

To date, the precise mechanisms through which financial education might promote positive changes in financial behavior remains a nascent area of financial literacy literature (Peeters et al., 2018). Our more holistic approach to assessing financial education gains supports the stance of Lyons and Neelakantan (2008) that the development (and assessment) of financial education has lacked a coherent theoretical approach. For one, separating objective from subjective financial knowledge would seem a particularly critical aspect when computing any form of composite financial literacy score. To illustrate, had we used a simpler evaluation approach involving only an objective knowledge-based assessment, it would have been difficult to delineate between the two positive states of *Knowledgeable* and *On Top of It*. We observe that critical differences between these two states in terms of sign and/or strength of association come in terms of financial attitudes and psychological characteristics—*Financial Self-Efficacy, Propensity to Plan for Money,* and *Financial Risk Tolerance*. A simpler evaluative approach, then, would be expected to indicate a lower ceiling of improvements for groups performing well, missing out on additional means by which education could further improve individuals' financial lives

through positively influencing their attitudes and traits.

Getting financial education right remains a perennial issue for policymakers, and is the subject of renewed interest around the world. In the U.K., recent initiatives underline younger people as a critical cohort (Money and Pensions Service, 2020). These initiatives are driven in part by a desire to increase the sub-standard levels of financial literacy in students (OECD, 2016). A further issue concerns the heterogeneity with which financial education programs are evaluated, and their impacts reported (Miller et al., 2015). Our Latent Transition Analysis approach to classifying characteristic financial literacy groups highlights that a segmented approach to evaluation that specifies cohorts into nominal groups of financial literacy would offer solutions to aforementioned policy challenges by (i) encouraging the adoption of a standardized set of financial literacy "states", and (ii) ensuring that education initiatives are tailored such that participants in each state receive particular focus on the domain of financial literacy known to be deficient for that state, whether it be regarding their objective or subjective financial knowledge, financial attitudes, or beneficial psychological characteristics.

6.1. Limitations and future research

Firstly, the fact that our sample only comprises students at a WEIRD (Western, Educated, Industrialized, Rich and Democratic) university may limit international generalizability (Henrich et al., 2010). Secondly, student samples can deviate from general public samples with respect to psychological and attitudinal variables (Hanel and Vione, 2016), and so attempts to expand our work should look to broaden the types of individuals studied. Thirdly, despite having a control group, it was not possible to randomize allocation to the treatment and control groups. It is likely that those taking up the offer of a financial education initiative had an inherent interest in the topic. However, we included covariates of the starting states to control for factors driving the uptake of the financial education unit. Fourthly, as we restrict analysis to those with surveys at both points in time participant dropout is of concern. Gerrans (2021) investigated predictors of dropout when examining change in objective knowledge over time using earlier cohorts of students from the same unit examined here and found no observable characteristics predicted dropout. Nevertheless, future work can explore the impact of attrition by including dropout as an additional absorbing state (Pandolfi et al., 2023). Finally, the financial behavioral measures were self-reported, which raises potential issues of recall accuracy (Rosenman et al., 2011).

Despite these limitations, our study contributes to the literature by being the first to utilize a Latent Transition Analysis to assess heterogeneity in the effectiveness of a financial education unit, while using a broader conceptualization of financial literacy, going beyond objective financial knowledge to also include subjective financial knowledge, positive financial attitudes, and beneficial psychological characteristics. Our findings are important to incorporate in public policy, given prior work arguing that financial education programs cannot be evaluated using a "one-size-fits-all" approach (Lyons and Neelakantan, 2008, p. 111) and individuals' ever increasing responsibility for making consequential decisions affecting their immediate and future financial well-being (Hoffmann and Plotkina, 2020).

CRediT authorship contribution statement

Paul Gerrans: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Arvid O.I. Hoffmann:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Simon J. McNair:** Writing – review & editing, Writing – original draft, Formal analysis. **Jason I. Pallant:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pacfin.2025.102669.

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